

Freight Rate Prediction — Assessment Report

Machine Learning Engineer Assessment · Spotter

1. Train / Test Validation Approach

The labeled development data (data/train-test.csv, 48,000 loads) was used in full for model development. Rather than carving out a single fixed hold-out split, model performance was estimated with 5-fold K-Fold cross-validation (shuffle=True, random_state=42). Each candidate model was trained and scored across all 5 folds, and the reported R^2 is the mean (with standard deviation) across folds rather than a single train/test split.

This approach was chosen over a single hold-out split for two reasons: (1) with ~48,000 rows, 5-fold CV gives a far more stable estimate of generalization error than one split, since every row is used for both training and validation across the folds; and (2) the low standard deviation of CV R^2 across folds (≤ 0.014 for every model tested) confirmed the data was homogeneous enough that fold-to-fold variance was not masking overfitting.

The 12,000-row data/validation.csv set was kept completely separate from training and model selection at all times — it was only ever passed through the final, already-tuned model to generate the required validation_predictions.csv. No labels exist for this set, so it could not be used for scoring; its role in the workflow is purely as the final inference set.

For hyperparameter tuning of the selected model (Ridge Regression), RandomizedSearchCV was run with the same 5-fold KFold splitter, searching alpha over a uniform range of 1–1001 across 50 sampled settings, again scored on R^2 .

2. Data Exploration & Quality Findings

Key findings

- posted_rate (target) ranges from \$57 to \$25,533, mean \$2,374 / median \$2,031, and is right-skewed.
- Distance and weight both show a clear positive relationship with posted_rate; equipment type (Reefer > Dry Van/Flatbed on average) and market_index are also informative.
- No duplicate rows or duplicate load_id values were found in the training data.
- Date range is consistent: 2025-01-01 to 2025-10-31 for training data; validation data covers November 2025.

Data-quality issues identified and how they were addressed

- weight: 300 missing values (0.63%) — imputed with the median weight within the same equipment type, so the imputed value reflects typical load weight for that equipment rather than a single global number.
- market_index: 374 missing values (0.78%) — imputed with the overall training-set median, since market_index did not show a strong date-level pattern that would justify a more granular imputation.
- Reported distance vs. geography: a Haversine great-circle distance was calculated from pickup/delivery lat-lon pairs and compared against the reported distance column (distance_ratio) as a sanity check and feature — reported distance is road distance and is expected to run higher than straight-line distance, and this ratio was used as a feature rather than a red flag.
- Unseen cities in validation: pickup/delivery city encoding was fit on the union of training and validation cities so no city is unseen at inference time; any genuinely unseen category would fall back to an explicit unknown code (-1) rather than erroring.

3. Feature Engineering

- Temporal: month, day-of-week, day-of-year, quarter, plus cyclical sin/cos encodings of month and day-of-week to capture seasonality without discontinuities.
- Geographic: Haversine calculated_distance, distance_ratio (reported vs. calculated), and pickup/delivery midpoint coordinates.
- Interactions: distance $\times$ weight, distance per unit weight, market_index $\times$ distance, quote_signal $\times$ distance.

- Location-pair aggregates: mean/median historical posted_rate, average distance, and average weight for each pickup–delivery pair, computed on training data and merged into both training and validation (missing pairs backfilled with the training median).
- Categorical encoding: one-hot encoding for equipment type; ordinal encoding for pickup/delivery city, fit across the union of train and validation cities.

4. Model Selection

Six regression approaches were trained and compared under identical 5-fold cross-validation:

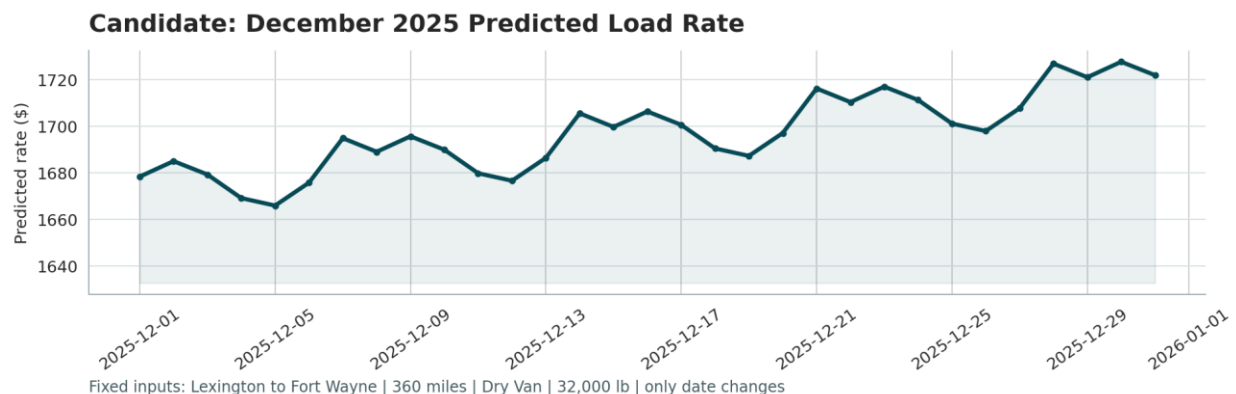
Model	CV R ² (mean)	CV R ² (std)	Train MAE	Train RMSE
Ridge Regression (tuned)	0.8514	0.0012	\$150.42	\$572.77
Lasso Regression	0.8502	0.0137	\$147.79	\$575.39
LightGBM	0.8463	0.0126	\$118.67	\$517.42
XGBoost	0.8428	0.0121	\$111.81	\$472.30
Gradient Boosting	0.8386	0.0134	\$112.91	\$473.51
Random Forest	0.8314	0.0139	\$85.23	\$370.06

Ridge Regression was selected as the final model. Although the tree-based ensembles (Random Forest, Gradient Boosting, XGBoost, LightGBM) fit the training data more tightly (lower training MAE/RMSE), their cross-validation R² was equal to or slightly below tuned Ridge, indicating the extra training-set fit was not translating into better generalization. Ridge also trains in a fraction of a second versus minutes for Random Forest/Gradient Boosting, and its coefficients are directly interpretable — pair_rate_mean, distance, and market_distance_interaction were the strongest positive drivers of predicted rate, while pair_avg_distance, calculated_distance, and pair_rate_median contributed negatively, consistent with the location-pair aggregate features capturing most of the lane-specific pricing signal.

Final tuned Ridge Regression performance (5-fold CV): R² ≈ 0.851, MAE ≈ \$150.42, RMSE ≈ \$572.77.

5. December 2025 Fixed-Lane Prediction Chart

The chart below is produced by the provided score.py logic against the completed december-chart-inputs.csv (one row per day, December 1–31, 2025, all other inputs fixed at Lexington → Fort Wayne, 360 miles, Dry Van, 32,000 lb). Only the date varies across rows, so the chart isolates the model's learned seasonal/day-of-month pricing signal on an otherwise identical lane.



Predicted rates for the fixed Lexington–Fort Wayne lane rise gradually across December, from roughly \$1,678 on December 1 to roughly \$1,728 by December 30, with day-to-day fluctuation layered on top of that trend. This is consistent with the model's temporal features (cyclical month/day-of-week encodings) picking up a mild seasonal upward drift heading into peak winter shipping, without any of the fixed lane inputs changing.

6. Deliverables Summary

- `validation_predictions.csv` — `load_id`, `predicted_rate` for all 12,000 rows in `data/validation.csv`.
- `december-chart-inputs.csv` — the provided template filled with `predicted_rate` for all 31 December days.
- `best_model.pkl` / `scaler.pkl` — the trained Ridge Regression model and its `StandardScaler`.
- `freight_rate_ml.py` / `Freight_Rate_ML.ipynb` — full pipeline: exploration, cleaning, feature engineering, training, tuning, prediction, and validation/scoring.